Daeun Lee

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RESEARCH INTERESTS

- **Multimodal (Video) Reasoning/Generation/Understanding** with RL and post-training
- **Embodied AI with Sensor Modalities** like Gaze, Audio, 3D LiDAR, HD Maps etc

EDUCATION

The University of North Carolina at Chapel Hill — NC, United States
Ph.D. Computer Science (Advisor: Mohit Bansal) — *Aug. 2024 —*

Korea University — Seoul, South Korea
B.E. Department of Statistics — *Mar. 2019 — Feb. 2024*

RESEARCH EXPERIENCES

Google XR — CA, United States
Student Researcher (Mentor: Sagnik Majumder) — *Dec.2026 —*

Meta Superintelligence Labs — CA, United States
Research Intern (Mentor: Markus Woodson, Tobias Hinz) — *May.2026 — Dec.2026*

Adobe Research — CA, United States
Research Intern (Mentor: Subhojyoti Mukherjee, Branislav Kveton) — *May.2025 — Nov.2025*

UNC Chapel Hill — NC, United States
Research Assistant (Advisor: Mohit Bansal) — *Aug.2024 — Current*

KAIST — Seoul, South Korea
Research Intern / Contract Researcher (Advisor: Sung Ju Hwang) — *Mar.2023 — Aug.2024*

NAVER LABS — Jungja, South Korea
Research Intern (Mentor: Minheok Heo) — *Jul.2022 — Dec.2022*

Korea University — Seoul, South Korea
Research Intern (Advisor: Jinkyu Kim) — *Jul.2021 — Dec.2022*

PUBLICATIONS

[P2] IndustryNav: Exploring Spatial Reasoning of Embodied Agents in Dynamic Industrial Navigation
Yifan Li, Lichi Li, Anh Dao, Xinyu Zhou, Yicheng Qiao, Zheda Mai, Daeun Lee, Zichen Chen, Zhen Tan, Mohit Bansal, Yu Kong.
Under review

[C8] StreamGaze: Gaze-Guided Temporal Reasoning and Proactive Understanding in Streaming Videos
Daeun Lee, Subhojyoti Mukherjee, Brano Kveton, Ryan Rossi, Viet Lai, David Seunghyun Yoon, Franck Dernoncourt, Trung Bui, Mohit Bansal.
The 40th Neural Information Processing Systems, Evaluations and Datasets Track ***(NeurIPS), 2026.***
Also accepted to CVPR 2026 with strong scores (5/5/5) but desk-rejected after camera-ready because of coauthor's reviewing policy issue.

[C7] VisionCoach: Reinforcing Grounded Video Reasoning vis Visual-Perception Prompting
Daeun Lee, Shoubin Yu, Yue Zhang, Mohit Bansal.
The 19th European Conference on Computer Vision ***(ECCV), 2026.***

[C6] Self-Correcting Text-to-Video Generation with Misalignment Detection and Localized Refinement
Daeun Lee, Jaehong Yoon, Jaemin Cho, Mohit Bansal.
The 64th Annual Meeting of the Association for Computational Linguistics ***(ACL) Findings, 2026.***

[C5] Video-Skill-CoT: Skill-based Chain-of-Thoughts for Domain-Adaptive Video Reasoning
Daeun Lee*, Jaehong Yoon*, Jaemin Cho, Mohit Bansal.
Empirical Methods in Natural Language Processing ***(EMNLP) Findings, 2025.***

[C4] BECoTTA: Input-dependent Online Blending of Experts for Continual Test-Time Adaptation
Daeun Lee*, Jaehong Yoon*, Sung Ju Hwang.
*International Conference on Machine Learning (**ICML**), **2024***

[C3] HD Maps are Lane Detection Generalizers: A Novel Generative Framework for Single-Source Domain Generalization
Daeun Lee, Minhyeok Heo, Jiwon Kim.
*CVPR Data-Driven Autonomous Driving Simulation Workshop (**CVPRW**), **2024***

[C2] Resolving Class Imbalance for LiDAR-based Object Detector by Dynamic Weight Average and Contextual Ground Truth Sampling
Daeun Lee, Jinkyu Kim.
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), **2023***

[C1] Bridging the Domain Gap towards Generalization in Automatic Colorization
Hyejin Lee, Daehee Kim, Daeun Lee, Jinkyu Kim and Jaekoo Lee.
*European Conference on Computer Vision (**ECCV**), **2022***

[P1] Trajectory Prediction by Clustering Human Interactions at Multiple Scales
Chiho Choi*, Daeun Lee*, Srikanth Malla, Sangjae Bae, Jinkyu Kim.
Preprint

ACADEMIC SERVICES

Reviewer

CVPR 2022-2026, ECCV/ICCV 2022-2026, AAAI 2025, NeurIPS 2025-2026

Invited Talks

Sep 2025. **SKKU** (**Sungkyunkwan University**) Google Developer Groups .
Jan 2025. **Cisco**. *'Reliable Text-to-Video Generation'*

AWARDS/HONORS

NeurIPS 2026 Travel Award *Oct.2026*

ICML 2024 Travel Award *Jun.2024*

Digital Innovation Big Data Contest *May.2021*
2nd place, *Korea Enterprise Data Corp.(KED)*

ICT Autonomous Driving Project *Dec.2020*
5st place, *The Federation of Korean Information Industries*

Financial Big Data Festival *Dec.2020*
1st place, *MiraeAsset.Corp*

Kakao Arena Competition *May.2020*
Top 2%, *Kakao.Corp*

Outstanding Student Scholarship *2021-2023*
Korea University

REFERENCES

Mohit Bansal *Professor at UNC Chapel Hill* mbansal@cs.unc.edu

Jaemin Cho *Professor at Johns Hopkins University* jaemin@jhu.edu

Branislav Kveton *Principal Research Scientist at Adobe Research* kveton@adobe.com

Markus Woodsen *Research Engineer at Meta Superintelligence Labs* markusw@meta.com

Tobias Hinz *Research Scientist at Meta Superintelligence Labs* thinz@meta.com

Sung Ju Hwang *Professor at KAIST* sjhwang82@kaist.ac.kr